



TASK 1 REPORT
LAB SETUP REPORT

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TASK: TASK 1
DURATION: DAY 1 – DAY 12
DATE: SEPTEMBER 2026

1. INTRODUCTION

The purpose of this task was to establish a basic cybersecurity laboratory environment for performing security testing and learning fundamental cybersecurity concepts. The laboratory was created using virtual machines so that security tools and testing activities could be performed in a controlled and isolated environment.

Kali Linux was configured as the attacker and security-testing machine, while Metasploitable2 was configured as the intentionally vulnerable target machine. The two virtual machines were connected through an isolated internal network named **cyberlab**. Basic Linux commands, networking commands, cryptographic operations and cybersecurity tools were also practiced as part of the environment setup.

2. OBJECTIVE

The main objectives of this task were:

- ❖ To understand the fundamentals of cybersecurity.
- ❖ To set up a virtual cybersecurity laboratory.
- ❖ To install and configure Kali Linux.
- ❖ To configure Metasploitable2 as a vulnerable target machine.
- ❖ To establish secure communication between the virtual machines.
- ❖ To practice fundamental Linux commands.
- ❖ To understand basic networking concepts and perform connectivity testing.
- ❖ To perform basic cryptographic operations using OpenSSL.
- ❖ To become familiar with cybersecurity tools such as Nmap, Wireshark, Burp Suite and Netcat.

3. LAB ENVIRONMENT

The cybersecurity laboratory was created using Oracle VirtualBox. Two virtual machines were used:

COMPONENT	PURPOSE
Kali Linux	Attacker and security-testing machine
Metasploitable2	Intentionally vulnerable target machine
VirtualBox	Virtualization platform
cyberlab	Isolated internal network

The laboratory was configured so that security testing could be performed without exposing the intentionally vulnerable target machine to an external network.

4. KALI LINUX SETUP

Kali Linux was installed and configured in VirtualBox as the attacker machine. Kali Linux provides various security-testing and network-analysis tools required for cybersecurity practice.

The virtual machine was configured with appropriate memory, processor and networking resources. The first network adapter was configured using NAT for normal connectivity, while the second adapter was connected to the isolated cyberlab internal network.

```
(kali@kali)~[/cybersecurity]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 80539sec preferred_lft 80539sec
    inet6 fd17:625c:f037:21be2:10e1:6928:59dd/64 scope global dynamic noprefixroute
        valid_lft 86146sec preferred_lft 14146sec
    inet6 fe80::ada1:cd7e:63c7:ba03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:33:a0:cf brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 scope global eth1
        valid_lft forever preferred_lft forever
(kali@kali)~[/cybersecurity]
$
```

5. METASPLOITABLE2 SETUP

Metasploitable2 was configured as the target machine for the cybersecurity laboratory. It is an intentionally vulnerable Linux virtual machine designed for security testing and educational purposes. The machine was created in VirtualBox using the existing Metasploitable2 virtual disk. Its network adapter was connected only to the isolated cyberlab internal network.

The target machine was configured with the IP address: 192.168.100.20

```

RX bytes:5576 (5.4 KB)  TX bytes:6032 (5.8 KB)
Base address:0xd020 Memory:f0200000-f0220000

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:5e:a7:7c
          inet addr:192.168.100.20  Bcast:192.168.100.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe5e:a77c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1080 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1162 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:78050 (76.2 KB)  TX bytes:92080 (89.9 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:602 errors:0 dropped:0 overruns:0 frame:0
          TX packets:602 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:274137 (267.7 KB)  TX bytes:274137 (267.7 KB)

msfadmin@metasploitable:~$
```

6. NETWORK CONFIGURATION AND CONNECTIVITY TEST

An isolated Internal Network named cyberlab was created in VirtualBox to allow communication between Kali Linux and Metasploitable2. The network configuration used in the laboratory was:

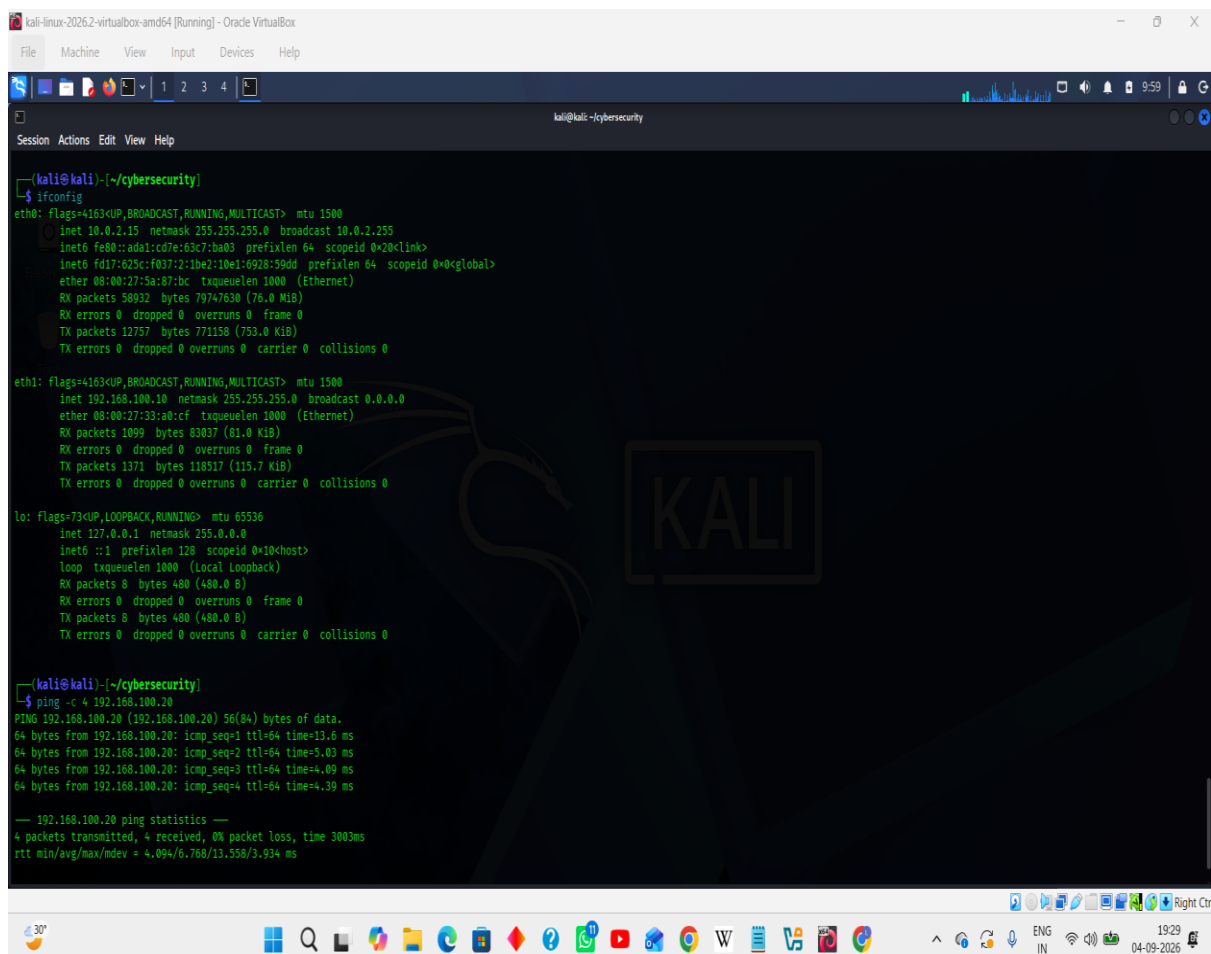
Kali Linux used its eth1 interface to communicate with Metasploitable2 through the isolated internal network. The IP configuration of Kali Linux was verified using the ip addr command.

MACHINE	INTERFACE	NETWORK	IP ADDRESS
Kali Linux	eth1	cyberlab	192.168.100.10
Metasploitable2	eth0	cyberlab	192.168.100.20

Connectivity between the Kali Linux attacker machine and the Metasploitable2 target machine was tested using the ping command.

The following command was used: ping -c 4 192.168.100.20

The test successfully received responses from the target machine, confirming that communication between Kali Linux and Metasploitable2 was working correctly over the isolated cyberlab network.



```
kali-linux-20262-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~/cybersecurity

kali@kali:~/cybersecurity$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::ada1:c7e:63c7:ba03 prefixlen 64 scopeid 0<20<link>
    inet6 fd17:625c:f037:2:1be2:10e1:6928:59dd prefixlen 64 scopeid 0<0<global>
    ether 08:00:27:5a:87:bc txqueuelen 1000 (Ethernet)
    RX packets 58932 bytes 79747630 (76.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12757 bytes 771158 (753.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.100.10 netmask 255.255.255.0 broadcast 0.0.0.0
    ether 08:00:27:33:a0:cf txqueuelen 1000 (Ethernet)
    RX packets 1099 bytes 83037 (81.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1371 bytes 118517 (115.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

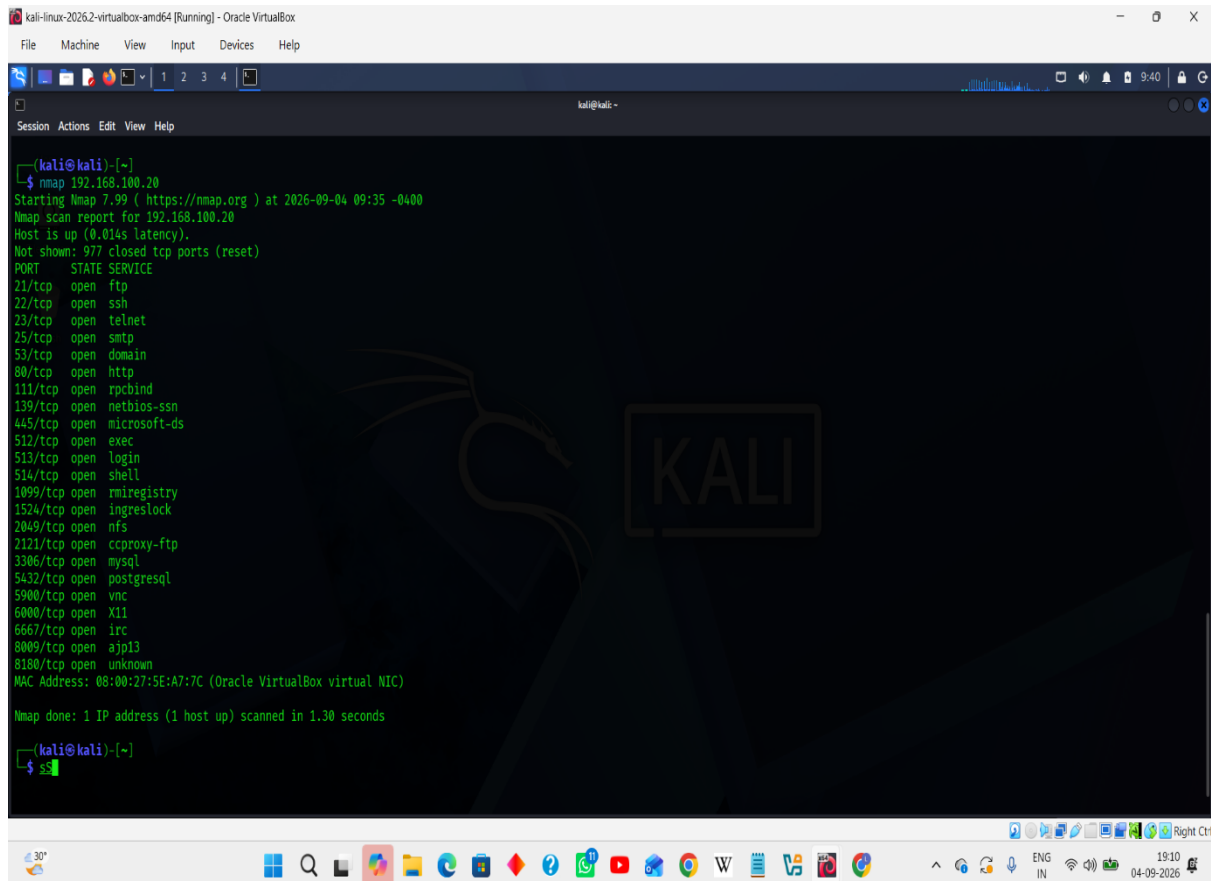
kali@kali:~/cybersecurity$ ping -c 4 192.168.100.20
PING 192.168.100.20 (192.168.100.20) 56(84) bytes of data:
64 bytes from 192.168.100.20: icmp_seq=1 ttl=64 time=13.6 ms
64 bytes from 192.168.100.20: icmp_seq=2 ttl=64 time=5.03 ms
64 bytes from 192.168.100.20: icmp_seq=3 ttl=64 time=4.09 ms
64 bytes from 192.168.100.20: icmp_seq=4 ttl=64 time=4.39 ms

--- 192.168.100.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 4.094/6.768/13.558/3.934 ms
```

7. NMAP NETWORK SCANNING

Nmap was used from Kali Linux to scan the Metasploitable2 target and identify available network services. The following command was executed: `nmap 192.168.100.20`

The scan confirmed that the target machine was active and identified several open services. Some of the identified services included FTP, SSH, Telnet, HTTP, MySQL, PostgreSQL and VNC. This demonstrated the use of Nmap for basic host discovery and service enumeration within the controlled laboratory environment.



```
kali@kali:~$ nmap 192.168.100.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-04 09:35 -0400
Nmap scan report for 192.168.100.20
Host is up (0.014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:5E:A7:7C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.30 seconds

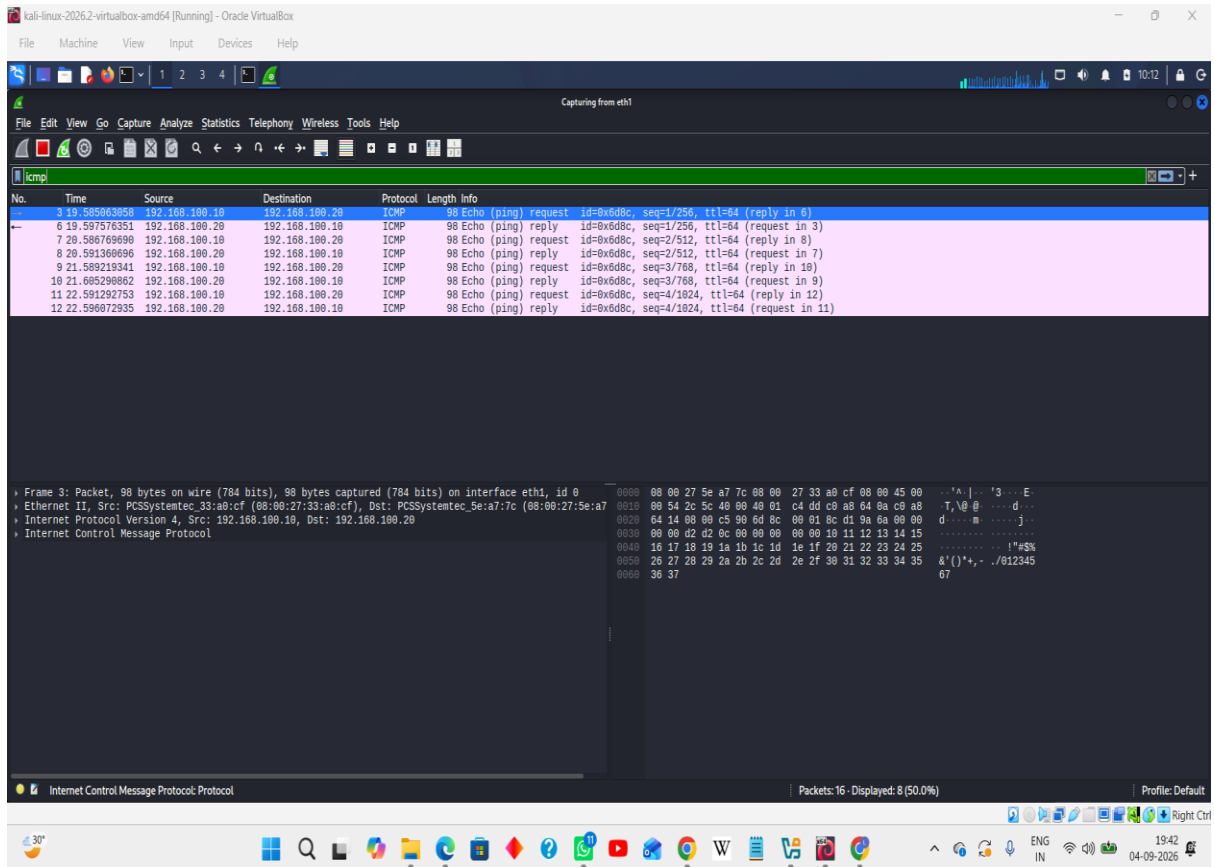
kali@kali:~$
```

8. WIRESHARK PACKET CAPTURE

Wireshark was used to capture and analyse network packets generated during communication between Kali Linux and Metasploitable2.

An ICMP ping was performed between the two machines, and the captured traffic was filtered using: `icmp`

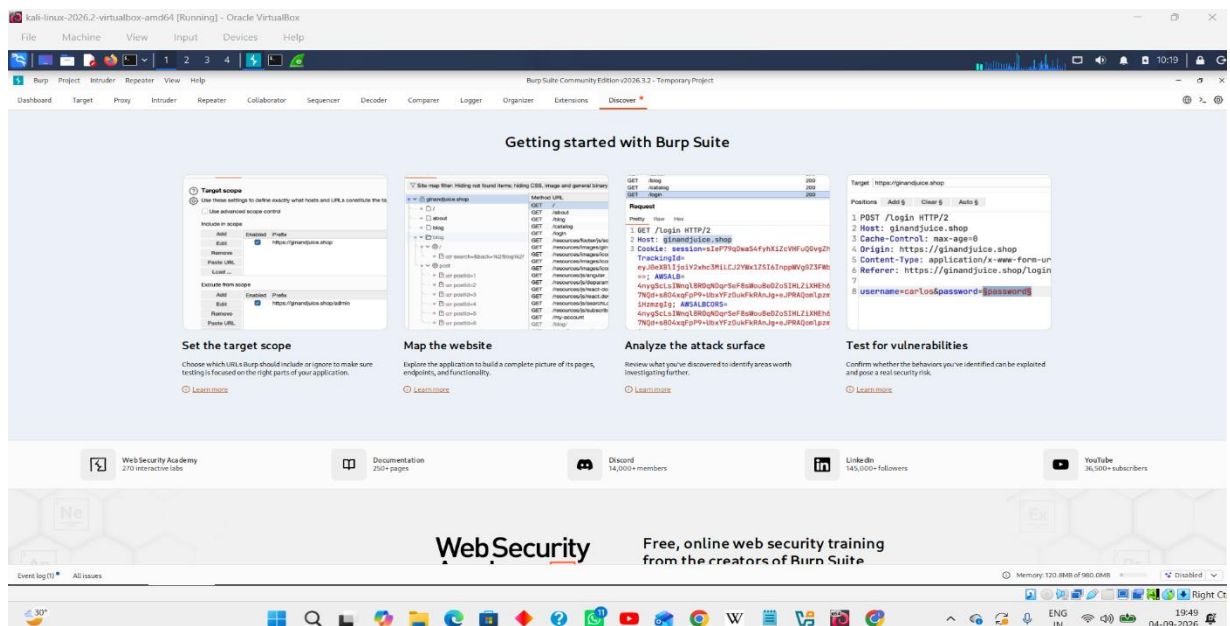
The filter displayed the ICMP request and reply packets generated during the connectivity test.



9. BURP SUITE

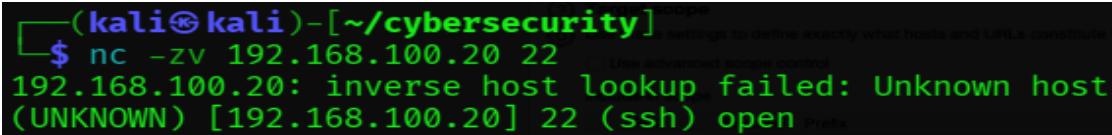
Burp Suite was explored as part of the security-tool familiarization activity.

Burp Suite is a web security testing platform that can be used to inspect and analyse HTTP/HTTPS requests and responses. The Burp Suite Community Edition interface was launched successfully in the Kali Linux environment.



10. NETCAT

Netcat was used to test network connectivity to an individual service on the Metasploitable2 target. The following command was used: `nc -zv 192.168.100.20 22`. The result showed that port 22 (SSH) was open on the target machine. This demonstrated how Netcat can be used to verify the availability of a specific network service.



```
(kali㉿kali)-[~/cybersecurity]
$ nc -zv 192.168.100.20 22
192.168.100.20: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.100.20] 22 (ssh) open
```

11. RESULTS AND OBSERVATIONS

The cybersecurity laboratory was successfully configured using Kali Linux and Metasploitable2. Both machines were connected through the isolated **cyberlab** network, and communication between them was successfully verified.

Nmap successfully identified multiple open services on the Metasploitable2 target. Wireshark successfully captured ICMP traffic generated during the connectivity test. Basic Linux commands and cryptographic operations were also successfully performed. Burp Suite and Netcat were launched and used for basic tool familiarization.

The practical activities provided hands-on experience with the basic tools and techniques required for further cybersecurity testing.

12. CONCLUSION

The cybersecurity laboratory environment was successfully established using VirtualBox, Kali Linux and Metasploitable2. The isolated network configuration enabled safe communication between the attacker and target machines.

Through this setup, fundamental Linux commands, networking concepts, cryptographic operations and cybersecurity tools were practiced. Nmap, Wireshark, Burp Suite and Netcat were successfully explored as part of the initial cybersecurity environment setup. This laboratory provides a controlled foundation for performing further cybersecurity learning and security-testing activities.